

Designing LED lighting products

Engineers involved in designing innovative and affordable LED lighting solutions can now benefit from a broad product portfolio of LED drivers, alternating current (AC)/direct current (DC) and DC/DC power management devices, wireless and wired interface control and embedded processors. They have the option of not only controlling the power stage but regulating LED currents as well, eliminating the need for multiple components and reducing the system cost.

A very significant part of LED lighting is the driver design. "The reliability of the driver is a crucial factor to achieve longer life expectations. While designing the driver, it is important to pay attention to the selection of components, which can give a longer lifetime," says Apoorva Awasthy, business development manager-analogue, Texas Instruments India, which also offers a suite of LED design tools.

LED lighting systems can be designed to accurately control voltage and current regulation for precise LED light intensity and colour mixing, temperature monitoring to prevent thermal runaway, intelligent and adaptive dimming of the LED, and fault detection (over voltage or current, blown string). Communication with external systems is also possible via power-line communication (PLC), wireless technology or interfaces.

York City, every New Year's Eve) being the well-known ones.

Cherreddi recollects, "A rather interesting application of LEDs in recent times has been in the areas of mood-based and architectural lighting. A great example of that is at NYC Waterfalls, where a series of manmade waterfalls installed around New York Harbour have been lit up using LED lighting fixtures to recreate the effect of moonlight on water. The application of LEDs in media, entertainment, outdoor displays, TV and cinema is at an early stage—this century will be lit up by LED, just as the incandescent bulb largely lit up the last."

Driving home with LED

LEDs have been used for the amber and red taillights in cars for quite some time now, but as far as forward-lighting goes, it has been discussed for long but not implemented, as the efficiency of LEDs was insufficient for headlights thus far. Prototypes have been demonstrated since 2006, and hobbyists have tweaked their cars to fit special LED headlights. However, it's only now that LED headlights are showing signs of becoming part and parcel of mainstream car models. Headlight maker Hella is working on at least 13 models of car headlights with LEDs.

"With the AUDI A8, it was demonstrated now for the first time how this advanced feature can be realised in a

volume car. We expect this contribution of LED technology to leave the level of high-end vehicles and reach the middle-class segment in reasonable time," hopes Reichl.

While car headlights are usually 12V applications, the US military seems to have even started using 24V LED headlights for their trucks.

LED key for mobile projection

Reichl points out, "An application that cannot be realised with other

light sources (except lasers) is one of the more consumer-oriented applications—projection from mobile devices (phones or cameras) with embedded or companion projectors. High-performance LEDs are the key to this application, with high efficiency of the top emitting LED chips, the small size, long lifetime and high product quality."

Other than miniaturising mobile projectors, LEDs have also helped in blowing up screens. Flatscreen displays with LED backlights are all the rage now. The image quality is much better when LCD TVs use LEDs as the light source instead of fluorescent lamps. LED TVs are more environment-friendly, as these consume less power and use no mercury in their manufacture. LEDs are also small and produce less heat, so these can be housed in thinner cases—the reason why you can now find really flat screens (only about 5cm thin) and even flexible displays.

"The advent of light-emitting diodes (LED) technology could be counted as one of the major innovations in the flat-screen display space. LED-based flat screens provide marvellous visual appeal, lower energy consumption, longer lifetime, improved robust-



OSRAM LED light points for dipped beam in Audi A8